AI ACADEMY · LEVEL 5 · GRADE 10 · WEEK 15 · TEACHER GUIDE

Training and validation curves

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

Make each curve value and model choice auditable.

Show complete pairs,known split differences and frozen-state evaluations;separate an evaluated checkpoint from training or final evidence.

Keep one habit: Assess independent loss columns and restored choice with checked keys;retain exact-tie and comparison conventions.

MAIN LESSON · SECTION 1

Prepare the connected mathematical investigation

> Keep explanation,shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Compute comparable **training** and validation losses at saved **checkpoint**s

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Printed lesson and workbook; paper and pencil

Calculator and optional local Python 3 editor; no packages or AI accounts needed

Use the fictional examples supplied; prepare the fresh retry before class

Training loss measures fit to the examples used for updates. Independent performance needs suitable separate **evidence**, and settings selected using validation must not be presented as independent final testing.

Opening retrieval

Without opening the old lesson, explain one key idea from Learning rate experiments. Give an example and a mistake that the idea helps you avoid.

Look for: Run a controlled comparison of learning rates

Reference example, not the only acceptable wording: Teacher explanation: First v = 0.5 with MSE 5.625. Second gradient -7.5 gives v = 0.875 and MSE 3.1640625. Distance is multiplied by 0.75 each step.

Without opening the old lesson, explain one key idea from **Training** a small network. Give an example and a mistake that the idea helps you avoid.

Look for: Trace repeated forward loss gradient and update steps

Reference example, not the only acceptable wording: Teacher explanation: **Prediction**s [0, 0], errors [-2, -4], MSE 10, gradient -10 and new v = 1.

Without opening the old lesson, explain one key idea from Slopes and local change. Give an example and a mistake that the idea helps you avoid.

Look for: Calculate an average rate of change over a stated interval

Reference example, not the only acceptable wording: Teacher explanation: Average slope is (0 - 4) / 2 = -2. The local slope at 2 is zero. They describe different locations or intervals.

Pictured end goal

By the end, I can explain **training** and **validation curve**s using a traced example and a stated limitation.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Learning rate experiments

Training a small network

Slopes and local change

For the compact alternative schedule, use two 45-minute sessions. Have students draw **checkpoint** number horizontally and MSE vertically, using one clearly labelled scale for both curves.

Explicitly name the differing target rules in this synthetic example so the picture is not misrepresented as proof of a single over**fitting mechanism**. Keep **checkpoint** selection separate from parameter updates.

Outcomes

Compute comparable **training** and validation losses at saved **checkpoint**s

Choose among observed **checkpoint**s using a declared validation rule

Distinguish development **evidence** from final test **evidence**

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Teach frozen-state evaluation,selection and restoration

> Keep main five-checkpoint data and pictured three-epoch log distinct.

Main configuration,paired checkpoint3calculation and guided key

Keep the restricted network h = ReLU(x), y = vh, with only v trained. **Training** pairs remain (1, 2) and (2, 4). Start v = 0 and make four full-data updates at rate 0.1. Save states v = 0, 1, 1.5, 1.75 and 1.875.

At each saved state, evaluate both splits without changing v during evaluation.

For this deliberately constructed demonstration, validation pairs are (3, 4.5) and (4, 6). Their targets follow a different slope from the **training** pairs.

This makes the two trends disagree visibly; it is not **evidence** that these few rows represent a real population. Validation labels do not enter the gradient.

Both columns use ordinary MSE in the same target units, evaluated on their own fixed rows. **Training** MSE is 2.5(v - 2) squared; validation MSE is 12.5(v - 1.5) squared.

Different **input** values and target rules affect the loss levels. A gap alone does not identify its cause.

For **checkpoint** 3, v = 1.75. **Training prediction**s are [1.75, 3.5], residuals [-0.25, -0.5], and MSE 0.15625. Validation **prediction**s are [5.25, 7], residuals [0.75, 1], and MSE 0.78125.

All four **prediction**s must use the same saved value.

The code evaluates supplied **checkpoint**s; it does not train on validation data. It assumes finite numeric pairs and selects the lowest observed validation MSE, breaking exact ties in favour of the earlier **checkpoint**.

The rule is chosen before inspecting the values.

Complete checkpoint 3(v1.75)paired calculation · Input x · Target · Prediction vReLU(x) · Residual prediction–target · Squared error

> **Complete checkpoint 3(v1.75)paired calculation:** Training A
>
> **Input x:** 1
>
> **Target:** 2

> **Prediction vReLU(x):** 1.75
>
> **Residual prediction–target:** -0.25
>
> **Squared error:** 0.0625

> **Complete checkpoint 3(v1.75)paired calculation:** Training B
>
> **Input x:** 2
>
> **Target:** 4
>
> **Prediction vReLU(x):** 3.5
>
> **Residual prediction–target:** -0.5
>
> **Squared error:** 0.25

> **Complete checkpoint 3(v1.75)paired calculation:** Validation C
>
> **Input x:** 3
>
> **Target:** 4.5
>
> **Prediction vReLU(x):** 5.25
>
> **Residual prediction–target:** 0.75
>
> **Squared error:** 0.5625

> **Complete checkpoint 3(v1.75)paired calculation:** Validation D
>
> **Input x:** 4
>
> **Target:** 6
>
> **Prediction vReLU(x):** 7
>
> **Residual prediction–target:** 1
>
> **Squared error:** 1

The current program evaluates the five supplied saved vvalues; it does not run training or change vusing validation gradients. The training sequence can be reproduced from the earlier complete four-update experiment. Each saved state is held fixed while both training and validation MSEs are computed. Choose lowest observed validation MSE, then earliest index on an exact tie.

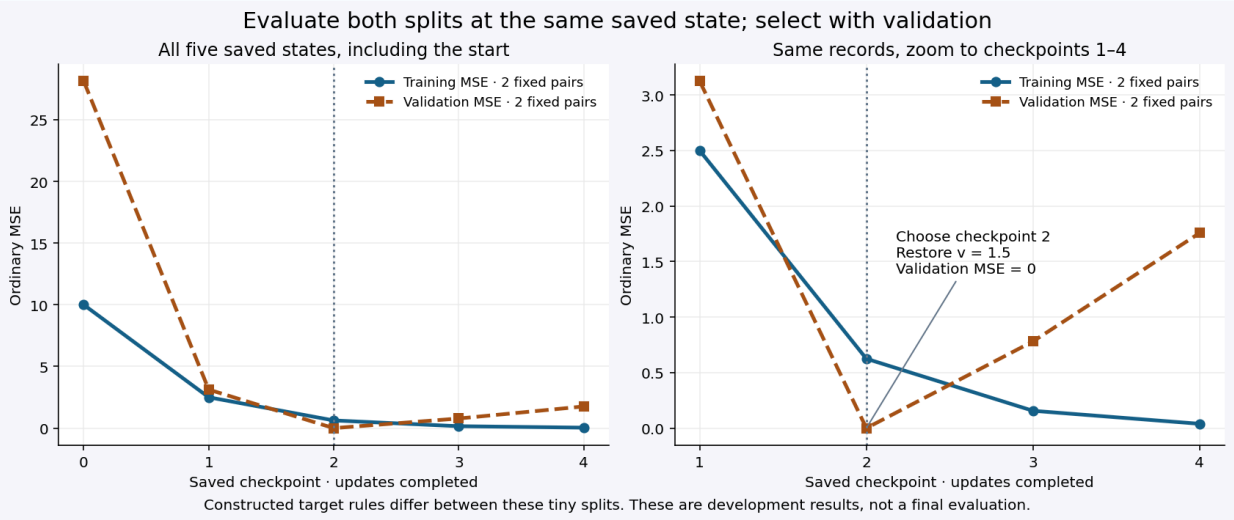
Restore the selected saved parameter v1.5, not the last loop variable v1.875. Zero loss on two constructed validation cases is limited model-selection evidence, not a final evaluation.

Guided: **training** changes by 0.0390625 - 0.625 = -0.5859375; validation changes by 1.7578125 - 0 = 1.7578125. Select **checkpoint** 2 with v = 1.5.

Exit: zero on selected validation examples is limited development **evidence**, and choosing with test results turns that test into development data.

Freeze choices using **training**/validation, then evaluate once on suitable untouched test examples and report the scope.

Complete numeric curves and labelled zoom



Separate pictured paper log and scope

Plot **training** and validation loss for **checkpoint**s before opening final cases.

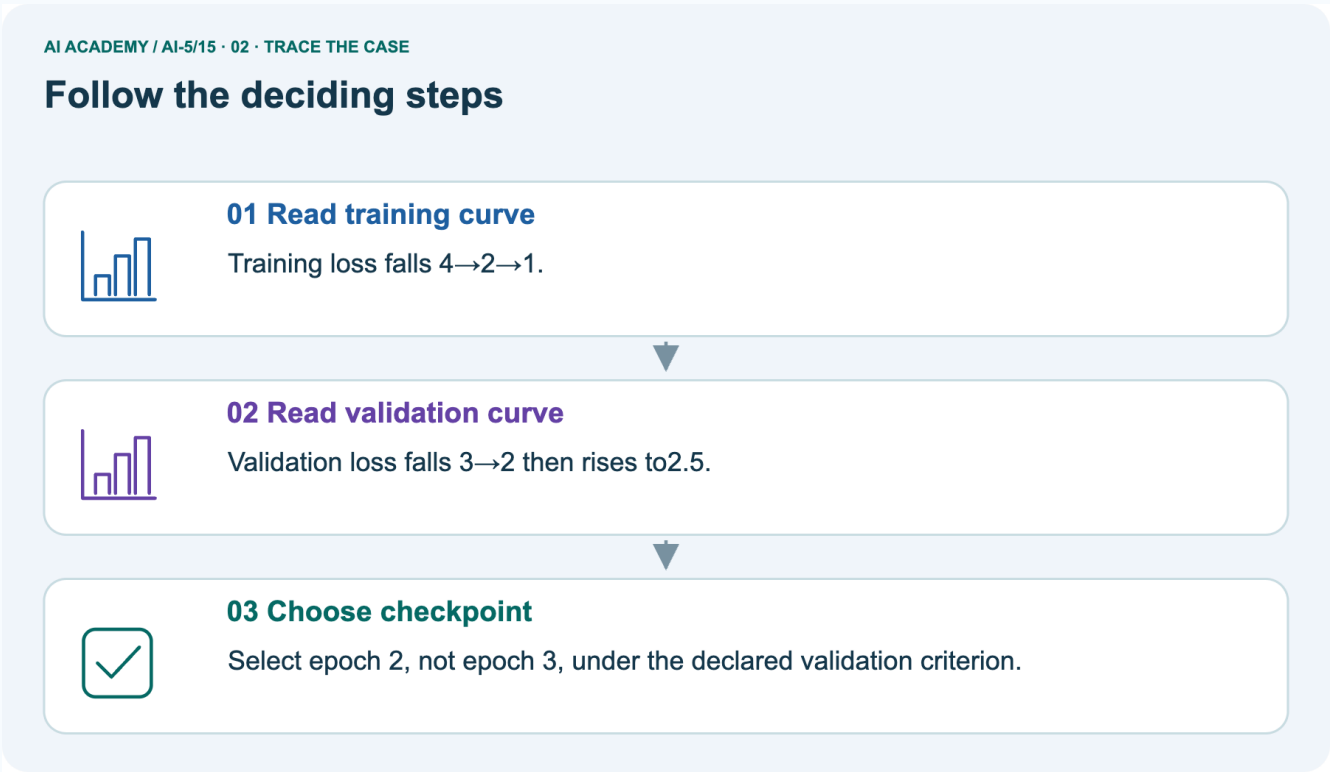
Supplied **checkpoint** losses: epoch 1 train 4, val 3; epoch 2 train 2, val 2; epoch 3 train 1, val 2.5. Selection: smallest validation loss.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The picture is a separate supplied three-epoch log:training 4/2/1, validation 3/2/2.5. Minimum validation chooses epoch 2, even though training is lowest at epoch 3. If only epoch 3 validation changes to1.5, the same rule instead chooses epoch 3. No training code or row-level predictions are supplied for that paper log.

The main executable five-checkpoint fixture includes all four numeric data pairs and chooses checkpoint 2, v1.5; its checkpoint count is not the pictured epoch numbering.

Trace the unchanged criterion under one changed value



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Read training curve	Training loss falls 4→2→1.
2. Read validation curve	Validation loss falls 3→2 then rises to2.5.
3. Choose checkpoint	Select epoch 2, not epoch 3, under the declared validation criterion.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

The **training curve** measures fit to examples used for learning. The **validation curve** measures a separate set used to choose a **checkpoint**. The final **training** loss is smallest here, but its validation loss has risen.

Under the supplied **selection rule**, keep **checkpoint** two. A later final test must remain outside both learning and **checkpoint** selection.

A single curve has sampling and run variability; validation-selected **checkpoint**s still require untouched final **evidence**.

Change: Epoch 3 validation loss is instead 1.5.

Prediction: Select epoch 3 under the unchanged criterion.

Reason: Its measured validation loss is now below epoch 2′s2; the **training curve** alone never settled selection.

▼ Earlier wording, alternatives and picture transcripts

Term	Meaning
Checkpoint	Saved model parameters and the information needed to identify a training state
Training curve	A record of a training measure at successive saved states
Validation curve	A record of a development measure on separate examples at those states
Selection rule	A declared procedure for choosing among candidate models or checkpoints

Later updates improve only one split

Series	2	3	4
Training solid line	0.625	0.15625	0.0390625
Validation dashed line	0	0.78125	1.7578125

This chart shows checkpoints 2 to 4 only, on one MSE scale from 0 to 2. Circles and a solid line mark training; squares and a dashed line mark validation. Lines connect measured checkpoints, not additional evaluations. The full table includes checkpoints 0 and 1.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Later updates improve only one split

Series	2	3	4
Training solid line	0.625	0.15625	0.0390625
Validation dashed line	0	0.78125	1.7578125

This chart shows checkpoints 2 to 4 only, on one MSE scale from 0 to 2. Circles and a solid line mark training; squares and a dashed line mark validation. Lines connect measured checkpoints, not additional evaluations. The full table includes checkpoints 0 and 1.

Using the table, calculate the training and validation loss changes from checkpoint 2 to checkpoint 4. Which saved parameter does the selection rule choose?

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Training and validation curves

The facts we will use

AI ACADEMY / AI-5/15

Checkpoint losses

	training	validation
epoch 1	4	3
epoch 2	2	2
epoch 3	1	2.5

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/15

Training and validation curves

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Read training curve

Training loss falls 4 -> 2 -> 1.



2. Read validation curve

Validation loss falls 3 -> 2 then rises to2.5.



3. Choose checkpoint

Select epoch 2, not epoch 3, under the declared validation criterion.

CHANGE ONE THING

Epoch 3 validation loss is instead 1.5.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: Observed validation MSE at **checkpoint**s 0 to 4 is [5, 2, 1, 1, 1.2]. Lowest loss wins with earlier exact ties. Which **checkpoint** is selected? What if **training** loss is lowest at **checkpoint** 4?

Teacher check: Teacher explanation: Select **checkpoint** 2. The tie at 3 does not replace the earlier **checkpoint**. Lower **training** loss at 4 does not change a validation-based rule.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for **Training** and **validation curve**s. Proposed explanation: “The last **checkpoint** is always best”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Choose by the stated validation criterion and restore the saved parameters.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Can the original epoch 3 **training** loss 1 justify claiming best generalisation? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

No. Original validation loss 2.5 exceeded epoch 2′s2. Freeze the selected **checkpoint** before final evaluation. Changed-case check: Select epoch 3 under the unchanged criterion.

Its measured validation loss is now below epoch 2′s2; the **training curve** alone never settled selection.

Changed-case reasoning: Select epoch 3 under the unchanged criterion. Its measured validation loss is now below epoch 2′s2; the **training curve** alone never settled selection.

Collect each learner′s explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide squared-error columns and labelled axes, allowing calculators. Extension: derive the two quadratic loss expressions and locate their different minima.

Explain why this constructed mismatch makes the validation optimum occur before the **training** optimum.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1 start to 7 minutes	Retrieve why training performance alone cannot select for independent performance.
7 to 18 minutes	Identify the two fixed splits and trace checkpoint 1; complete W1.
18 to 32 minutes	Calculate checkpoint 3 by hand and plot all supplied points on one MSE scale.
32 to 45 minutes	Run or trace the evaluation code, then use W2 to practise the exact tie rule.
Session 2 start to 8 minutes	Complete the guided comparison of checkpoints 2 and 4; identify the parameter to restore.
8 to 20 minutes	Use W3 and W5 to examine metric comparability and alternative causes of a widening gap.
20 to 35 minutes	Collect W4 independently with both loss columns and the explicit selection rule.
35 to 45 minutes	Use the exit question and fresh retry; require a distinction between selection and untouched final testing.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main choose index 2/v1.5. Supported2→4 trainchange−.5859375, validchange+1.7578125. Exacttieschooseearlier. W3SSE12/6=MSE 2 versusvalidation3. Independenttrain(1, 2)/(2, 4), valid(1, 1)/(2, 2):v1 train2.5/valid 0; v1.5 both.625; v2 train0/valid 2.5. Validationchoosesv1, trainingchoosesv 2; constructed target slopes differ. Keep numeric comparisons, criterion and restored model explicit; no untouched final claim.

Diagnose the current misconceptions

The last **checkpoint** is always best: Choose by the stated validation criterion and restore the saved parameters.

A gap proves one specific cause: Check split composition, metrics and evaluation procedures; this demonstration has deliberately different target rules.

Validation is untouched final **evidence**: Once used to choose settings or **checkpoint**s, it is development **evidence**.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new schema or calculation transfer

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: Use h = ReLU(x), y = vh, candidate v values 1, 1.5 and 2. **Training** pairs are (1, 2), (2, 4); validation pairs are (1, 1), (2, 2). Calculate both MSE columns and select by minimum validation MSE.

Explain the disagreement with selection by **training** loss.

Teacher explanation: For v = 1, train MSE 2.5 and validation MSE 0. For v = 1.5, both MSE values are 0.625. For v = 2, train MSE 0 and validation MSE 2.5. Validation chooses v = 1; **training** chooses v = 2.

These constructed splits favour different slopes.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Recover a plotted point

At v = 1, compute both validation **prediction**s, residuals and MSE on (3, 4.5) and (4, 6).

Teacher explanation: **Prediction**s [3, 4], residuals [-1.5, -2], squares [2.25, 4], MSE 3.125.

W2 Use the declared rule

Observed validation MSE at **checkpoint**s 0 to 4 is [5, 2, 1, 1, 1.2]. Lowest loss wins with earlier exact ties. Which **checkpoint** is selected? What if **training** loss is lowest at **checkpoint** 4?

Teacher explanation: Select **checkpoint** 2. The tie at 3 does not replace the earlier **checkpoint**. Lower **training** loss at 4 does not change a validation-based rule.

W3 Repair incompatible measures

A **training** report shows sum of squared errors 12 over six rows. Validation shows mean squared error 3 over two rows in the same target units. A learner says **training** error is four times larger. Repair the comparison.

Teacher explanation: **Training** MSE is 12 / 6 = 2. Compare 2 with validation MSE 3, not sum 12 with mean 3. The split sizes and examples still limit interpretation.

W5 Limit a curve diagnosis

Training loss falls while validation loss rises. Give two checks before declaring model complexity the cause, and explain why repeatedly selecting with validation makes it development **evidence**.

Teacher explanation: Possible checks include comparable loss definitions, unchanged pre**process**ing and evaluation settings, split representativeness, leakage and label quality.

Repeated selection uses validation outcomes to guide choices, so final evaluation needs separate untouched examples.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: **training** changes by 0.0390625 - 0.625 = -0.5859375; validation changes by 1.7578125 - 0 = 1.7578125. Select **checkpoint** 2 with v = 1.5.

Exit: zero on selected validation examples is limited development **evidence**, and choosing with test results turns that test into development data.

Freeze choices using **training**/validation, then evaluate once on suitable untouched test examples and report the scope.

W1 Recover a plotted point

W2 Use the declared rule

W3 Repair incompatible measures

W4 Evaluate new checkpoints independently

Use h = ReLU(x), y = vh, candidate v values 1, 1.5 and 2. **Training** pairs are (1, 2), (2, 4); validation pairs are (1, 1), (2, 2). Calculate both MSE columns and select by minimum validation MSE.

Explain the disagreement with selection by **training** loss.

Teacher explanation: For v = 1, train MSE 2.5 and validation MSE 0. For v = 1.5, both MSE values are 0.625. For v = 2, train MSE 0 and validation MSE 2.5. Validation chooses v = 1; **training** chooses v = 2.

These constructed splits favour different slopes.

W5 Limit a curve diagnosis

Independent assessment and fresh retry

W4 requires all six MSE values and the validation choice v = 1. Fresh retry: candidates v = 0.5, 1, 1.5 and validation pairs (2, 2), (4, 4). MSE values are 2.5, 0 and 2.5, so choose v = 1.

Ask what the result establishes and why it is not final test **evidence** after selection.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect,support and revisit the mathematics

> Use the saved evidence to choose the next step.

Project and next-week bridge

Add a labelled **training** and **validation curve**s **evidence** sheet to Image-pattern Network Investigation. Show how you can compute comparable **training** and validation losses at saved **checkpoint**s.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide squared-error columns and labelled axes, allowing calculators. Extension: derive the two quadratic loss expressions and locate their different minima.

Explain why this constructed mismatch makes the validation optimum occur before the **training** optimum.

Week 16 introduces a weight penalty and a precise early-stopping policy.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Delayed return and support record

Suggested return points in this level: ai-5/16, ai-5/18, ai-5/21. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	

Record	Teacher evidence
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Compute comparable **training** and validation losses at saved **checkpoint**s

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

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>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: Observed validation MSE at **checkpoint**s 0 to 4 is [5, 2, 1, 1, 1.2]. Lowest loss wins with earlier exact ties. Which **checkpoint** is selected?

What if **training** loss is lowest at **checkpoint** 4?

Reasoned practice answer: Teacher explanation: Select **checkpoint** 2. The tie at 3 does not replace the earlier **checkpoint**. Lower **training** loss at 4 does not change a validation-based rule.

Check every learner before discussion: Use the worked case for **Training** and **validation curve**s. Proposed explanation: “The last **checkpoint** is always best”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Choose by the stated validation criterion and restore the saved parameters.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Epoch 3 validation loss is instead 1.5. Require a **prediction** and a reason before comparing. Expected result: Select epoch 3 under the unchanged criterion.

Reason: Its measured validation loss is now below epoch 2′s2; the **training curve** alone never settled selection.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project **checkpoint**: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner′s explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

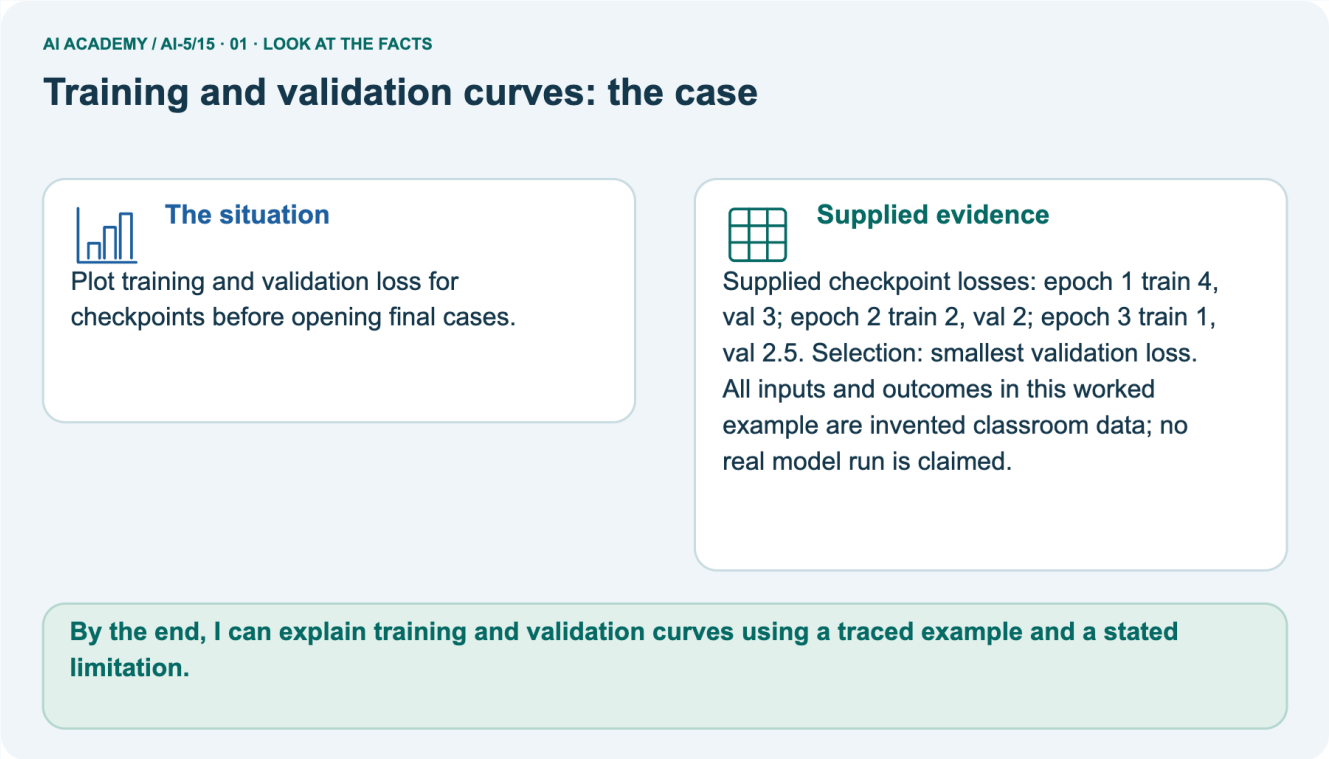
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Training and validation curves: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

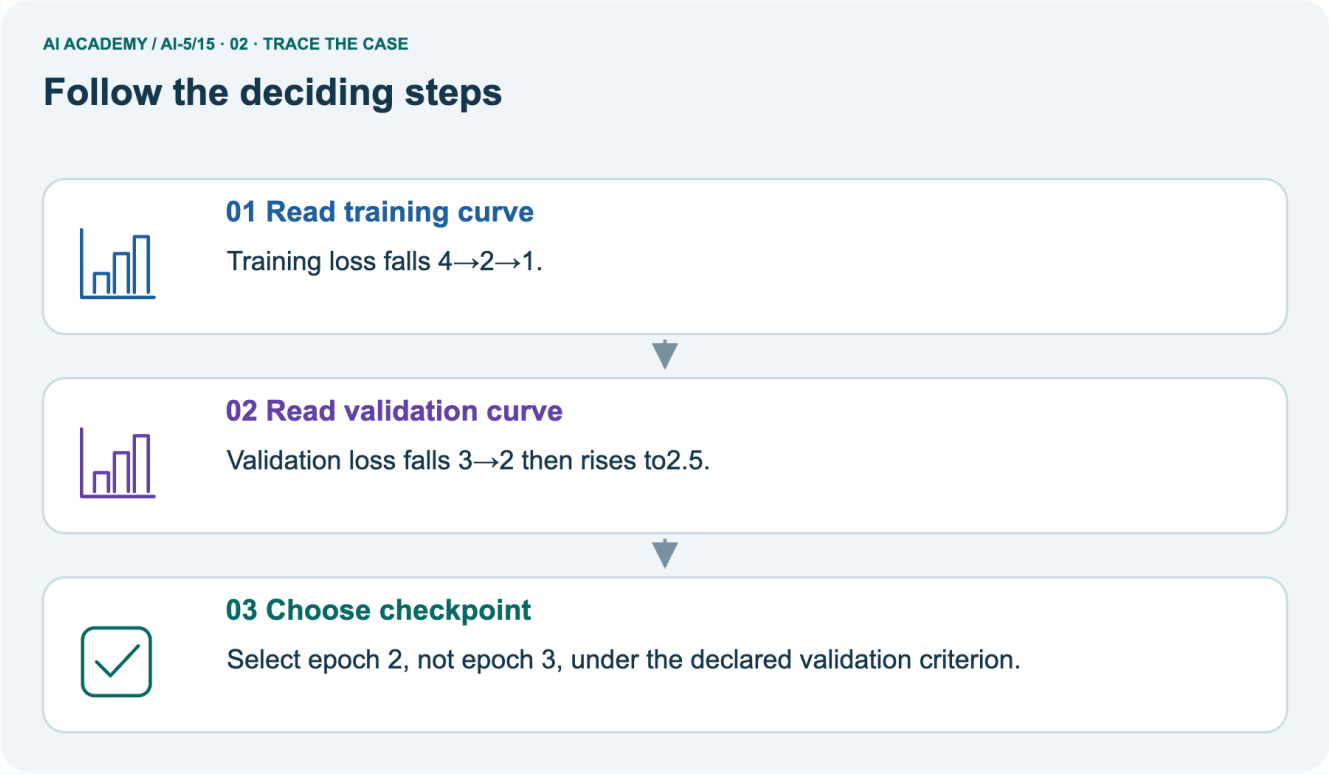
Plot **training** and validation loss for **checkpoint**s before opening final cases. Supplied facts: Supplied **checkpoint** losses: epoch 1 train 4, val 3; epoch 2 train 2, val 2; epoch 3 train 1, val 2.5. Selection: smallest validation loss.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Read **training curve**: **Training** loss falls 4→2→1. → Read **validation curve**: Validation loss falls 3→2 then rises to2.5. → Choose **checkpoint**: Select epoch 2, not epoch 3, under the declared validation criterion.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Checkpoint losses



Follow the labelled picture with the matching explanation. Open the vector diagram

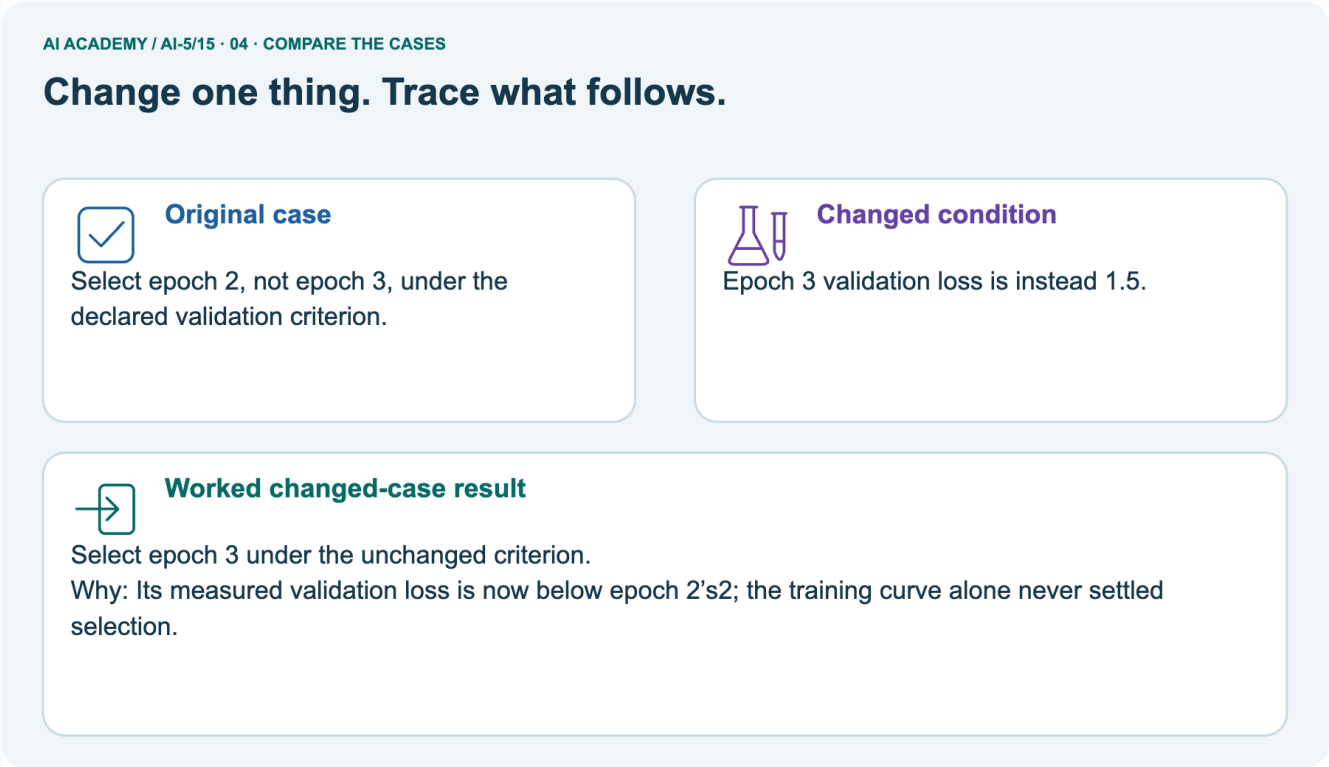
Checkpoint losses. epoch 1: **training** = 4, validation = 3. epoch 2: **training** = 2, validation = 2. epoch 3: **training** = 1, validation = 2.5.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Select epoch 2, not epoch 3, under the declared validation criterion. Changed condition: Epoch 3 validation loss is instead 1.5. Expected result: Select epoch 3 under the unchanged criterion.

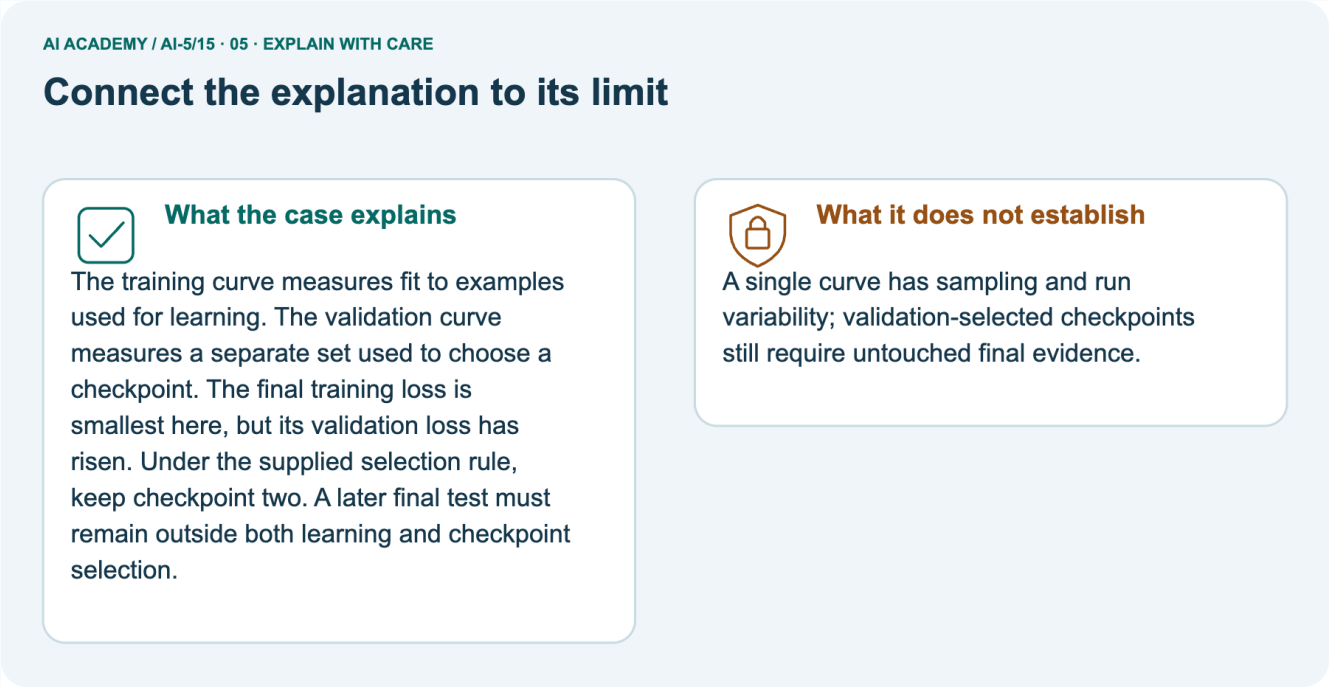
Reason: Its measured validation loss is now below epoch 2′s2; the **training curve** alone never settled selection.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Its measured validation loss is now below epoch 2′s2; the **training curve** alone never settled selection.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

The **training curve** measures fit to examples used for learning. The **validation curve** measures a separate set used to choose a **checkpoint**. The final **training** loss is smallest here, but its validation loss has risen.

Under the supplied **selection rule**, keep **checkpoint** two. A later final test must remain outside both learning and **checkpoint** selection.

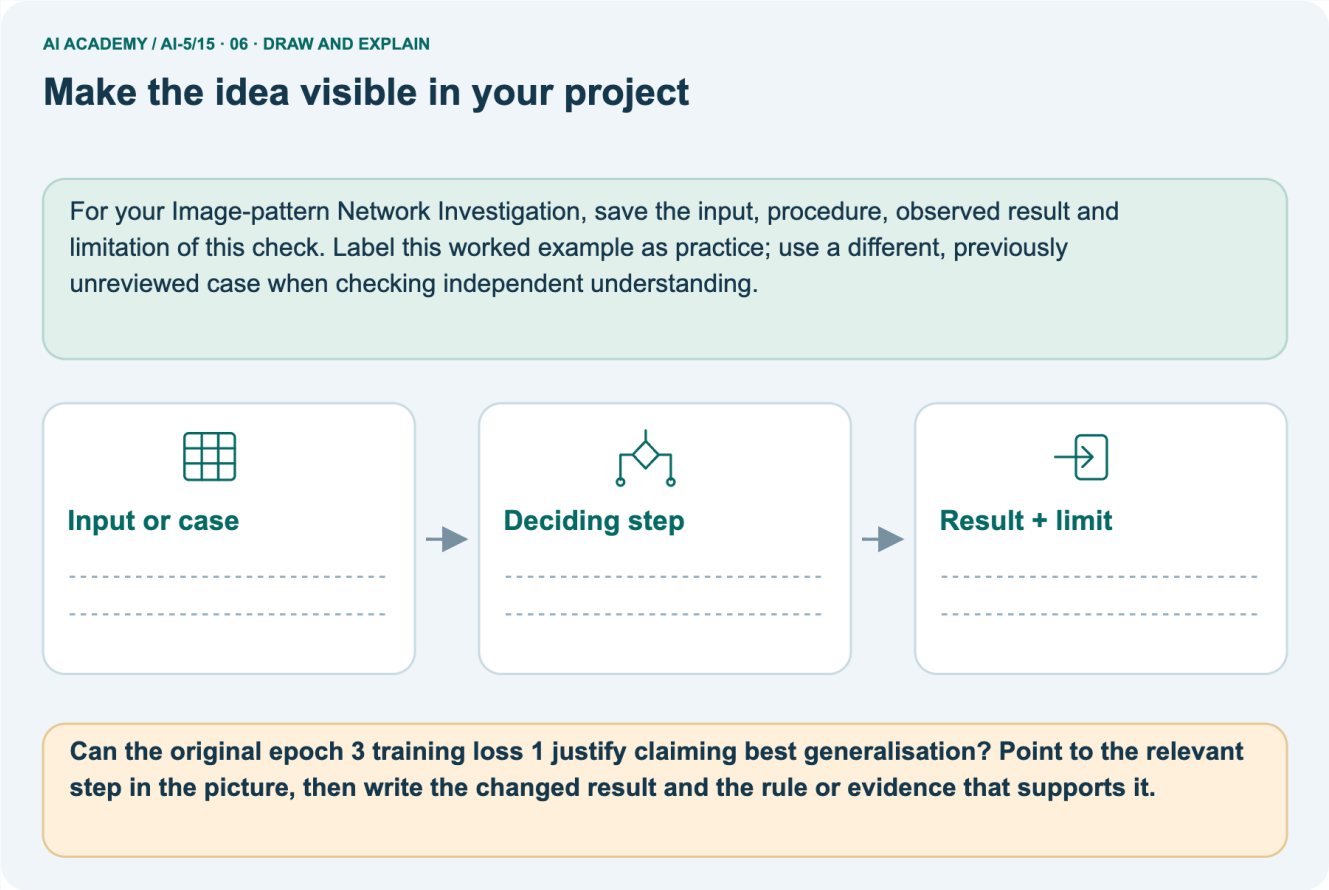
Limit: A single curve has sampling and run variability; validation-selected **checkpoint**s still require untouched final **evidence**.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: A single curve has sampling and run variability; validation-selected **checkpoint**s still require untouched final **evidence**.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Can the original epoch 3 **training** loss 1 justify claiming best generalisation? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: No. Original validation loss 2.5 exceeded epoch 2′s2. Freeze the selected **checkpoint** before final evaluation. Changed-case check: Select epoch 3 under the unchanged criterion.

Its measured validation loss is now below epoch 2′s2; the **training curve** alone never settled selection.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Training and validation curves: the case

02. Follow the deciding steps

03. Checkpoint losses

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.